

Abstract

Using MFC energy harvesters has become an attractive method to power sensor nodes in structural health monitoring systems of aircraft. This project aims to develop a finite element model able to aid the design and optimisation of potential MFC energy harvesting devices. This was achieved by carrying out an experimental procedure characterising two MFC energy harvesters by subjecting them to stress with an Instron tensile testing machine, and then using the data from these experiments to validate a model created in ANSYS mechanical APDL.

It was found that the produced finite element model was an accurate representation of the characteristics of the tested experimental devices, however, the simulated peak power output was higher than the experimental power, possibly due to energy losses in the system neglected in the model. This model should be useful in aiding the future designs of MFC energy harvesters.

Further research is required to validate the model over a range of different frequencies.

Keywords: MFC, model, composite, energy, harvest, aircraft

Table of contents

1.	Introduction and background.....	5
2.	Literature review.....	6
2.1.	Structural Health Monitoring.....	6
2.2.	Energy Harvesting.....	7
2.3.	Piezoelectric Energy Harvester Modelling.....	7
3.	Methodology and theory.....	8
3.1.	Experimental Method.....	8
3.2.	Modelling Method.....	9
3.2.1.	Model Geometry and Mesh.....	9
3.2.2.	Element Types.....	11
3.2.3.	Defining Material Properties.....	12
3.2.4.	FEM Configuration and Boundary Conditions.....	16
4.	Experimental Results and Permittivity Analysis.....	16
4.1.	Experimental Results.....	16
4.2.	Permittivity Analysis.....	18
5.	Presentation of Simulation Results.....	20
5.1.	Modelled Without Epoxy Bonding Layer.....	20
5.2.	Modelled Including Epoxy Bonding Layer.....	22
6.	Discussion and conclusions.....	23
7.	Project management, consideration of sustainability and health and safety.....	23
	References.....	28

1. Introduction and background

Energy harvesting and structural health monitoring of aircraft are heavily researched fields of engineering due to the potential financial and safety benefits. Maintenance and repairs of commercial aircraft make up roughly one quarter of operating costs^[1], and due to the increase in competition for commercial airliners, the average age of in-service aircraft is steadily increasing in an attempt to cut costs and increase profits^[2]. The reliability and safety of an aircraft decreases as it ages due to corrosion and fatigue, and inspections into the structural health of aircraft result in significant costs.

Structural health monitoring is a method of autonomous monitoring of fatigue and stress levels in aircraft structures, and involves integrating sensors within the structure of the aircraft. In order to minimize extra weight and wiring, and to negate the need for batteries, the sensors need to be self powered. Batteries have a finite lifespan. This means that they have to be either replaced or recharged regularly resulting in expensive aerospace maintenance work. Recently, using the piezoelectric effect to harness the energy of the surrounding environment has arisen as an attractive alternative to batteries^[3, 4, 5]. The strain energy present in the structure of the aircraft during flight can be harvested in order to power these sensors.

This project aims to develop a finite element model (FEM), scripted using ANSYS Mechanical APDL^[6], of a macro-fibre composite (MFC) based piezoelectric energy harvesting device (EHD) bonded to an orthotropic, layered composite. MFC is preferable to more traditionally used piezoelectric materials like lead zirconate titanate (PZT) due to its flexibility, allowing strain energy harvest under much larger deformations while maintaining the device's lifespan. The FEM script will be configurable with the capability to alter dimensions and properties in order to investigate the suitability of new strain energy harvester designs. This will make the FEM a useful model for validating new designs prior to manufacture which will help to reduce the costs associated with the testing and trial for the optimisation of EHD designs.

In order to validate the results of the FEM an experimental analysis is carried out to characterise the behaviour of two MFC EHDs. One of which consists of a composite cantilever beam bonded to an MFC patch with epoxy cured at 60psi, and the other cured in a

vacuum. The objective of the project is to validate the FEM by comparing the experimental values for voltage, current, and power with the simulated results.

2. Literature review

2.1. *Structural Health Monitoring*

Developments in micro-electro-mechanical systems (MEMS) technology at the start of the 21st century has allowed for low-power and low-mass sensor nodes capable of communicating over short distances^[7]. These small sensor nodes have limited computational ability but can be integrated into wireless sensor networks. An array of such sensors can transmit data pertaining to a specific property of the surrounding environment (for example strain, temperature, or vibrations) to a receiver node which can then store the data in a storage system to be processed.

These sensor networks are particularly useful for use in structural health monitoring (SHM) of aircraft^[8]. Here, the sensors can transmit data about the strain throughout the aircraft so that the operator can see if the aircraft has been used within its design limits. If these limits have been exceeded then it can be calculated how much this will decrease the lifespan of the aircraft, and if the vehicle has been operating below its design limit it can be calculated how the fatigue life could have increased. It is cost effective to exceed the design life of an aircraft, and with the current high levels of ageing aircraft in operation it is more important than ever to ensure their structural integrity^[2, 8].

Ideally the sensors need to be self powered as using the existing aircraft power supply causes problems with wiring and extra weight^[9]. The obvious solution to the problem appears to be to use batteries, but these have a finite lifespan and will need to be replaced or recharged, creating maintenance issues. There has been an abundance of research into different methods of energy harvesting to power sensor nodes, with many different proposed methods of converting free energy from the surrounding environment into usable electrical energy.

2.2. *Energy Harvesting*

Typical sensor nodes require a power supply of a few mW in order to properly function^[10], and this energy must be scavenged from the surrounding environment for use in aircraft

sensor nodes. Penella and Gassula^[11] present a review of available energy harvesters for powering sensor nodes. Possible methods of energy harvest include piezoelectric, thermoelectric, electromagnetic, solar, and biochemical methods. However, piezoelectric and thermoelectric methods are of most significance for aircraft SHM sensor networks.

Thermoelectric methods have been suggested^[12], but these require heavy heat storage units adding unnecessary weight to the aircraft and additional maintenance costs. Utilising the piezoelectric effect and strain energy harvesting using MFC is a more promising solution to the problem as it avoids these disadvantages.

MFC, or Macro Fibre Composite, is a piezoelectric material developed by NASA in 1996, an overview of MFC including many of its properties can be found on the manufacturers website^[13]. There has been a number of studies into the use of MFC as an energy harvester. Henry Sodano's work^[14] in which he compares MFC with the more traditional piezoelectric material PZT is of particular interest. In Sodano's work it is concluded that MFC is an inferior energy harvester, however, due to MFC's advantageous property of flexibility, an optimised MFC EHD would be very useful to the aerospace industry as it can be more easily bonded to curved structures such as aerofoils. More recently Guiliano et al.^[15] showed, by harvesting up to 12mW, that MFC EHD's do have the potential to be used in SHM systems.

2.3. Piezoelectric Energy Harvester Modelling

In 2009 Zhu et al.^[16] created a coupled piezoelectric-circuit FEM for the first time, capable of predicting power output and optimizing EHD design. Yang et al.^[17] created a FEM of an MFC EHD bonded to an aluminum cantilever beam, validated by results obtained from testing the device while under stress from an MFC actuator.

This project will build upon these previous developments by creating an FEM of an MFC based EHD bonded to a composite beam, validated by results obtained experimentally. The FEM will be configurable and will be able to assist in optimizing EHD designs for use with orthotropic, layered composite structures, which have different properties to the previously modeled isotropic aluminum structures. This will be of particular use for aircraft SHM systems as graphite/epoxy composites are increasingly becoming used as the material for aircraft structures due to their low density and high strength

3. Methodology and theory

In order to validate the results obtained from the model an experimental analysis was carried out first. This means that there were two main parts of this project, the experimental work and the modelling work.

3.1. *Experimental Method*

An experimental analysis was carried out on two different energy harvesters. The EHDs consisted of a layered IM6/950 composite beam with dimensions 53x300x5mm, bonded with Araldite 2014-1 epoxy to a Smart Materials M8528-P2 MFC patch, located in the centre of the substrate beam. The composite beam represents a structural element of an aircraft, and consisted of 32 laminae, each 156.25 μ m thick, with the fibres orientated in a symmetrical pattern which can be seen in figure 1 to the right.

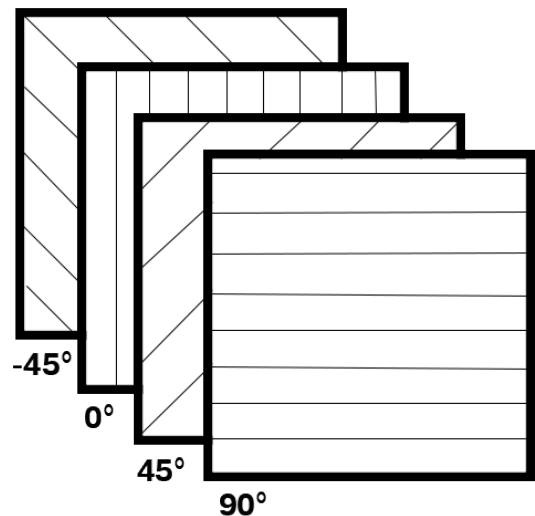


Figure 1: Illustration of the fibre layer orientation in the composite beam

One of the EHDs was made with epoxy cured in a vacuum, and the other was cured at 60psi. Using harvesters with different curing pressures allows the effect of the thickness of the epoxy bonding layer to be included and validated in the model.

An Instron tensile testing machine was used to vibrate the harvester at a frequency of 10Hz, with both sides of the harvester clamped. The force of vibration was determined by measuring the strain present in the substrate beam using a strain gauge, for both energy harvesters a strain of 250×10^{-6} was used, which corresponded to a force of 3kN for the vacuum cured harvester, and 2.9kN for the 60psi cured EHD.

A variable load resistor, R_L , was connected in series to the MFC patch, and the output voltage across R_L was measured using LabView^[18] and plotted for resistance values varying from

10k Ω to 200k Ω . The data output was a text file, and as the EHD produced an alternating current the data needed to be processed to find the rms voltage in order for the results to be useful. The data was processed in MatLab^[19] using the function rms. Using this method a value for the root mean squared output voltage for each R_L value was found. Using equation 1-1, Ohms law, the output rms current was calculated. Equation 1-2 was then used to calculate the output rms power produced by the energy harvester.

$$I = \frac{V}{R} \quad \text{Equation 1.1}$$

$$P = IV \quad \text{Equation 1.2}$$

Using this data the characteristics of the voltage, current and power outputs of the EHDs as R_L changes are defined, and can be used to validate the model.

3.2. Modelling Method

1.1.1. Model Geometry and Mesh

Ansys APDL was used to model the EHDs described in section 3.1. of this report. Figure 2 shows the geometry and co-ordinate system of the EHD used in the experiment, and thus the geometry created for use in the FEM. Table 1 shows the dimensions of the model, with the dimensions of the MFC obtained from the manufacturers data^[13].

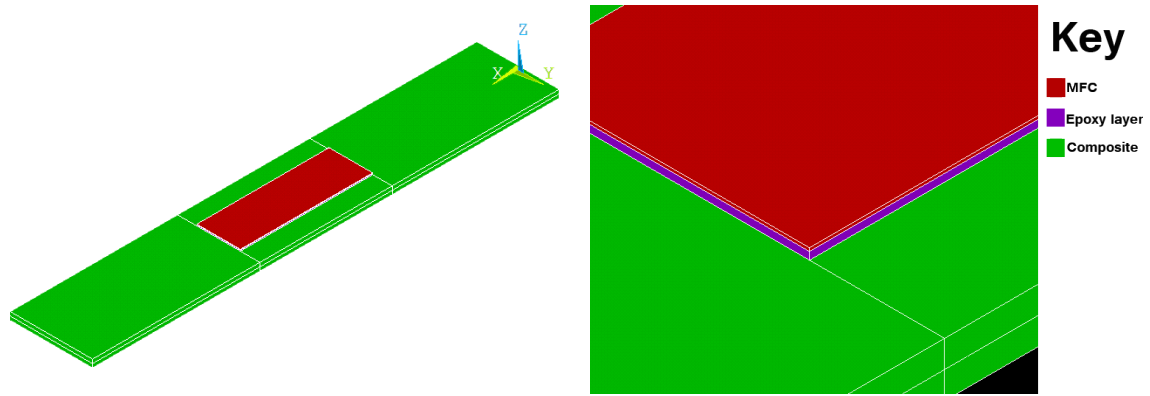


Figure 2: Model Geometry created in ANSYS with close up of epoxy layer

Dimension	Length (mm)
Composite substrate length	300
Composite substrate width	53
Composite substrate thickness	5
MFC length	85
MFC width	28
MFC thickness	0.3
Epoxy layer length	85
Epoxy layer width	28
Epoxy layer thickness	Variable

Table 1: Dimensions of the EHD

The epoxy bonding layer thickness is variable as the 60psi and vacuum EHDs used in the experiment have different epoxy layer thicknesses of 0.7mm and 1mm respectively.

Prior to extrusion of the geometry a 2D square mesh with an element size of 2.5mm was projected onto the 2D geometry. When extruded the composite beam was assigned a mesh division of 4, the epoxy and MFC layers were assigned a mesh division number of 2. The completed mesh can be seen in figure 3 overleaf.

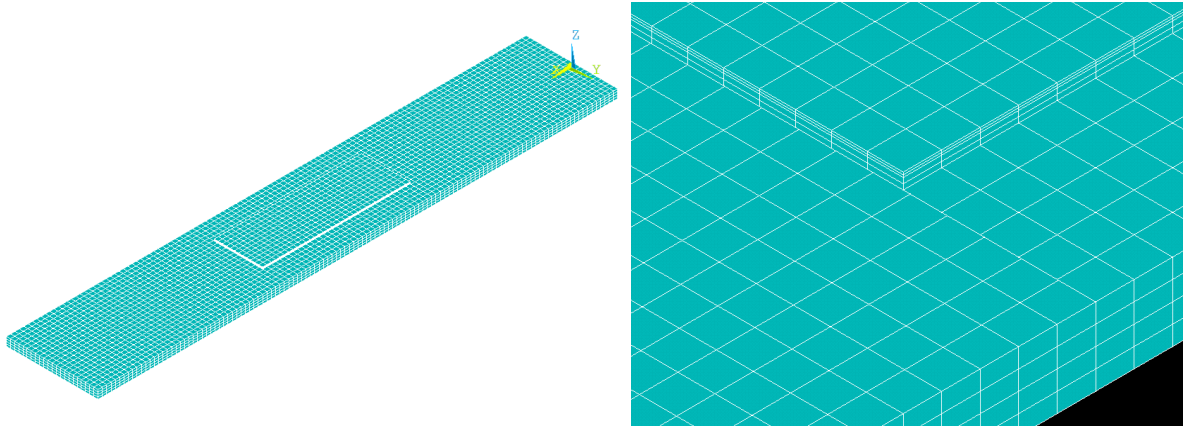


Figure 3: Diagram illustrating the mesh

1.1.2. Element Types

The different element types for the different sections of the model can be seen in table 2 below.

Section of Model	Element type used
MFC	SOLID5
Epoxy	SOLID45
Composite	SOLID185
Resistor	CIRCU94

Table 2: Element types used

The MFC layer is modelled using the SOLID5 element type. This is because the SOLID5 element type allows the coupling of mechanical and electrical fields, which is necessary for the MFC layer in this model. Electrodes were formed on the top and bottom of the MFC layer by grouping the nodes on either surface and coupling the electric potential degree of freedom across these nodes in order to keep the voltage profile constant. Using this method means that electrodes do not explicitly need to be modelled as real volumes. The electrodes are connected in series to the load resistor which is modelled using the element type CIRCU94.

The epoxy is being modelled as a linear isotropic material. In reality the epoxy characteristics vary with temperature and behave non-linearly, but for the purpose of analysing the power loss through the epoxy layer, as in this case, modelling it as isotropic is sufficient. SOLID45 is a suitable element type to use for the epoxy layer as it consists of 8 nodes and 3 degrees of freedom which will be compatible with both the SOLID5 elements used for the MFC layer,

and the SOLID185 elements used in the composite beam. SOLID185 was used for the composite as it has the capability to act as a layered structural element. This means that the fibre directions and layer thicknesses of the composite can be defined and modelled.

1.1.3. Defining Material Properties

In order to model the system the relevant material properties needed to be defined. For the MFC this meant that values for the elasticity matrix, $[c]$, piezoelectric stress matrix, $[e]$, relative permittivity, ϵ , and density, ρ , needed to be defined. And for the IM6/950 layered composite, the Young's modulus, E , Poisson's ratio, ν , and shear modulus, G , needed to be defined for each of the x , y , and z directions: as it is an orthotropic material. As well as this the orientation of the fibres in the different layers, and the thickness of the layers need to be defined for the composite.

Table 2 shows the relevant MFC material properties as stated on the manufacturers website^[13]. The subscripts indicate the orientation of the property with respect to the co-ordinate system. In this case, 3 is the poling direction.

Property	Value according to Smart Materials website ^[13]
E_{33}	30.336GPa
E_{11}	15.857GPa
ν_{31}	0.31
G_{31}	5.515GPa
d_{33}	-2.1E+02 pm/V
d_{31}	4.6E+02 pm/V
ρ	5440kg/m ³

Table 3: Relevant MFC properties available on manufacturers website

Using these values and equations 2.1 - 2.4, it is possible to calculate the elasticity matrix, $[c]$, and the piezoelectric stress matrix, $[e]$.

$$[e] = \begin{bmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \\ e_{14} & e_{24} & e_{34} \\ e_{15} & e_{25} & e_{35} \\ e_{16} & e_{26} & e_{36} \end{bmatrix} \quad [c] = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ c_{21} & c_{22} & c_{23} & c_{24} & c_{25} & c_{26} \\ c_{31} & c_{32} & c_{33} & c_{34} & c_{35} & c_{36} \\ c_{41} & c_{42} & c_{43} & c_{44} & c_{45} & c_{46} \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} & c_{56} \\ c_{61} & c_{62} & c_{63} & c_{64} & c_{65} & c_{66} \end{bmatrix}$$

Figure 4: The matrices of piezoelectric stress, $[e]$, and elasticity, $[c]$

MFC is transversely isotropic around the axis of the PZT fibres, therefore the compliance matrix, $[s]$, is as seen in equation 2.1^[20]. The compliance matrix is the inverse of the elasticity matrix, as shown in equation 2.2. The piezoelectric strain matrix, $[d]$, is calculated using equation 2.3. Because the applied electric field is in the direction of the poles the shear components d_{15} and d_{24} of the piezoelectric matrix are expected to be zero, d_{33} and d_{32} are also expected to be the same for MFC. $[d]$ can be converted to the piezoelectric stress matrix with equation 2.4.

$$s = \begin{bmatrix} \frac{1}{E_1} & \frac{-v_{31}}{E_1} & \frac{-v_{31}}{E_3} & 0 & 0 & 0 \\ \frac{-v_{31}}{E_1} & \frac{1}{E_1} & \frac{-v_{31}}{E_3} & 0 & 0 & 0 \\ \frac{-v_{31}}{E_1} & \frac{-v_{31}}{E_3} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{31}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \frac{(1+v_{31})}{E_1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \frac{(1+v_{31})}{E_1} \end{bmatrix} \quad [c] = [s]^{-1}$$

Equation 2-1

Equation 2-2

$$[d] = \begin{bmatrix} 0 & 0 & d_{31} \\ 0 & 0 & d_{32} \\ 0 & 0 & d_{33} \\ 0 & d_{24} & 0 \\ d_{15} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Equation 2-3

$$[e] = [c] \cdot [d]$$

Equation 2-4

By using equations 2.1 - 2.4 the values for the elasticity matrix, $[c]$, and the piezoelectric stress matrix, $[e]$ were calculated as shown below

$$[c] = \begin{bmatrix} 21.09 & 7.971 & 8.993 & 0 & 0 & 0 \\ 8.984 & 20.08 & 8.994 & 0 & 0 & 0 \\ 15.26 & 10.93 & 38.44 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5.516 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3.436 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3.436 \end{bmatrix} \times 10^9$$

$$[e] = \begin{bmatrix} 0 & 0 & 6.139 \\ 0 & 0 & -1.972 \\ 0 & 0 & -3.348 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Figure 5: The calculated elasticity matrix, $[c]$, and the piezoelectric stress matrix, $[e]$

In order to fully model the electromechanical coupling of the MFC EHD the relative permittivity of the active layer needed to be determined. Unfortunately there is a lot of disagreement in the literature as to what the value for the relative permittivity of MFC is, and this data is not provided by the manufacturer. A common method of estimating the permittivity is using mixing theory and estimating the MFC permittivity using the permittivity of PZT, the MFC epoxy, and the volume fraction of PZT in the active layer. As well as the compliance and stress matrices, the relative permittivity of MFC is required in order to simulate the behaviour of an MFC EHD. Estimates for the relative permittivity of MFC vary from 712 to 1694^[24, 25] compared with 3800 for PZT. This is a large variation and would vary the results of the simulation substantially. For this reason an analysis into the effect changing the value for the relative permittivity of the model has on the simulation results will be carried out in order to find the permittivity value that most closely correlates with the experimental data. This analysis can be found in section 4.2 of this report.

The epoxy used to bond the MFC to the composite in the experiment was Araldite 2014-1, its properties can be found in the manufacturers technical data sheet^[21] and are listed in table 4.

Material Property	Value for Araldite 2014-1 epoxy from datasheet ^[21]
Density	1600kg/m ³
Young's modulus	4GPa
Poisson ratio	0.35

Table 4: Epoxy properties

The final material properties that needed to be defined are the properties of the composite substrate beam. No data could be found for the exact IM6/950 composite that was used in the experiment, so instead values for a typical IM6/epoxy composite obtained from Amir Javidinejad's book "*Essentials of Mechanical Stress Analysis*"^[22] are used. These are shown in Table 5.

Material Property	Typical value for IM6/epoxy composite ^[22]
E_x	203GPa
E_y	11.2GPa
E_z	11.2GPa
G_x	8.4GPa
G_y	8.4GPa
G_z	8.4GPa
ν_{xy}	0.32
ν_{xz}	0.32

Table 5: properties of a typical IM6/epoxy composite

As well as the values displayed in table 5 a value for the Poisson's ratio of the 950 epoxy present in the experimental composite obtained from the manufacturer^[23], Cytec, was used as the value for ν_{yz} .

As well as defining the properties of the composite, the layer thickness and fibre orientation also needed to be defined. This was done in ANSYS mechanical APDL using SECDATA commands, and is possible due to the element type SOLID185 enabling the modelling of layered structural solids. A layer thickness of 156.25 μ m, and the layer orientation outlined in section 3.1 were defined.

1.1.4. FEM Configuration and Boundary Conditions

To simulate the force exerted by the Instron tensile testing machine in the experimental work a mechanical load in the form of $F=F_0\sin(2\pi ft)$ is applied at both ends of the composite plate in the x direction, at a force which results in a peak-to-peak micro strain equal to 250, as in the experiment. As well as this the nodes at either end of the plate are confined to only move in the x direction in order to simulate the clamp of the Instron machine.

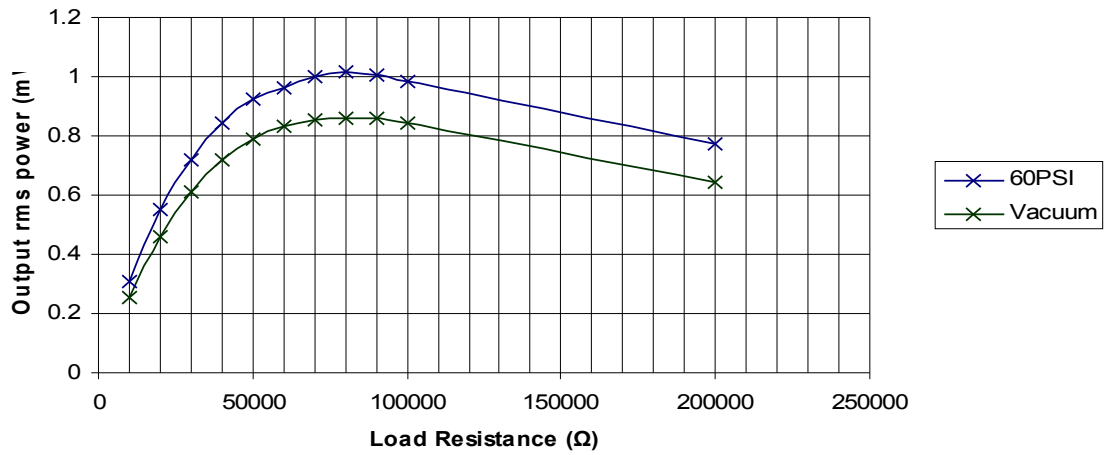
The simulation is run at a frequency of 10Hz for load resistance values between 10k Ω and 200 k Ω in steps of 10 k Ω . This is done in order to provide results that are easily comparable to the experimental results in order to validate the model.

The simulation is done initially with no epoxy layer modelled, and then with epoxy layer thicknesses of 0.7mm and 1mm, corresponding to the 60psi and vacuum experimental EHD epoxy layer thicknesses. By doing this the effect of adjusting the epoxy layer thickness can easily be visualised, and the simulation results can be compared to the experiment results.

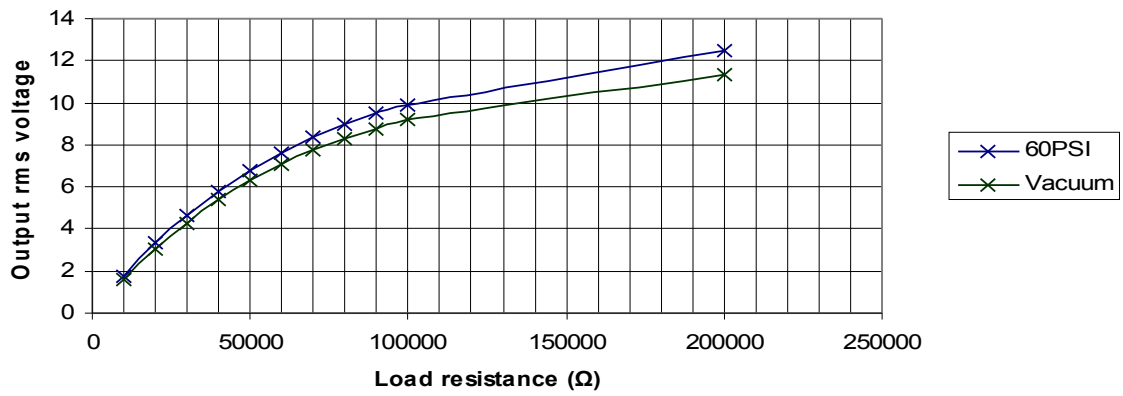
4. Experimental Results and Permittivity Analysis

4.1. Experimental Results

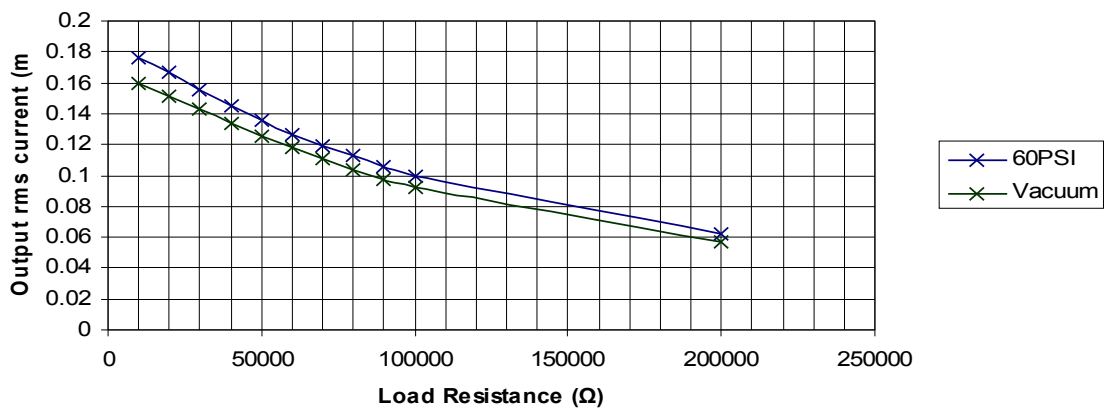
Graphs 1.1, 1.2 and 1.3 overleaf present the data collected from the experimental work, illustrating how the output rms power, voltage, and current vary as the resistance value of the load resistor is changed.



Graph 1.1: How power output varies with load resistance



Graph 1.2: How voltage output varies with load resistance



Graph 1.3: How current varies with load resistance

The behaviour of the EHDs can be easily understood by applying Kirchhoff's Law. At low values for the load resistance the output voltage tends to 0, and the EHD acts as a current source. Whereas at high R_L values, the current tends towards 0 and the EHD acts as a voltage source. Due to this, at both high and low R_L values the power is zero, and at a point between these two extremes is a maximum output power. In this case, Graph 1.1 shows that the R_L value where maximum output power occurs is $80k\Omega$, with an rms output power of around 1mW.

These graphs also show the power lost due to the thickness of the epoxy bonding layer. The EHD cured at 60psi, which has an epoxy layer thickness of 0.7mm, generates more power than the EHD cured in a vacuum, which has an epoxy layer thickness of 1mm. This suggests that, as expected, the thickness of the epoxy bonding layer contributes to power loss in the device, with a thicker layer of epoxy causing shear lag. This shear lag results in the strain of the MFC patch being lower than that of the composite plate due to the deformation of the epoxy, meaning a loss of power output. The power output difference caused by just 0.3mm extra epoxy demonstrates the importance of minimising the thickness of the epoxy bonding layer as much as possible when bonding these EHDs to aircraft structures.

The simulation should be able to replicate these results by showing the same general behaviour displayed by the experimental EHDs, as well as providing similar values for peak power output, and at what R_L value this occurs. The simulation should also be able to replicate the effect of increasing the thickness of the epoxy bonding layer.

4.2. *Permittivity Analysis*

A value for the relative permittivity of MFC is needed to correctly model its behaviour, and due to variation in the estimated values of the relative permittivity of MFC calculated in the literature, an analysis was carried out to examine the effect of varying the permittivity on the simulation results as compared to the experimental results. This was done in order to estimate a value for the relative permittivity of the MFC. The aim here is to find the permittivity value that results in the simulation matching the experimental results.

As expected the experimental results in section 4.1 of this report show that there is a maximum output power somewhere between the two extremes of $R_L=0$ and $R_L=\infty$. This maximum power occurs when the output impedance of the energy harvester is equal to the

impedance of the load resistor. As such, the value for R_L that will result in peak power is given by equation 3.1, shown below.

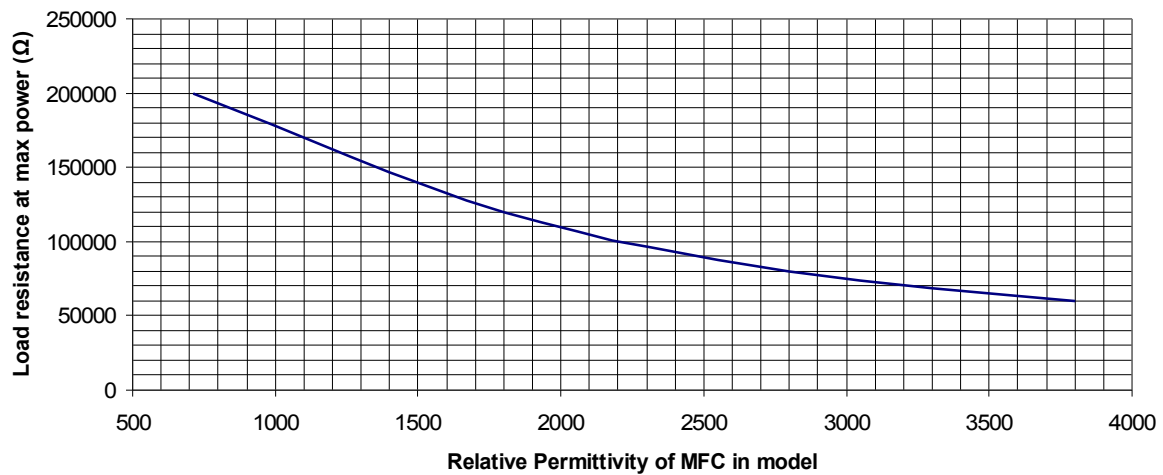
$$R = \frac{1}{2\pi fC} \quad \text{Equation 3.1}$$

Where C is the capacitance of the MFC layer, given by equation 3.1

$$C = \frac{\epsilon w l}{t} \quad \text{Equation 3.2}$$

Where ϵ is the relative permittivity of the MFC at constant strain, w is the width, l the length, and t the thickness.

Looking at equation 3.1 and 3.2 it is clear that increasing the value of ϵ should result in the peak power occurring at a lower load resistance, and as shown in the experimental results, in this case this should occur at $R_L=80k\Omega$. The FEM simulation was run several times using different values for ϵ , and the resulting values of the load resistor that provided maximum power output were plotted against ϵ , as seen in graph 2 below.



Graph 2: how changing ϵ in the FEM effects the optimum load resistance

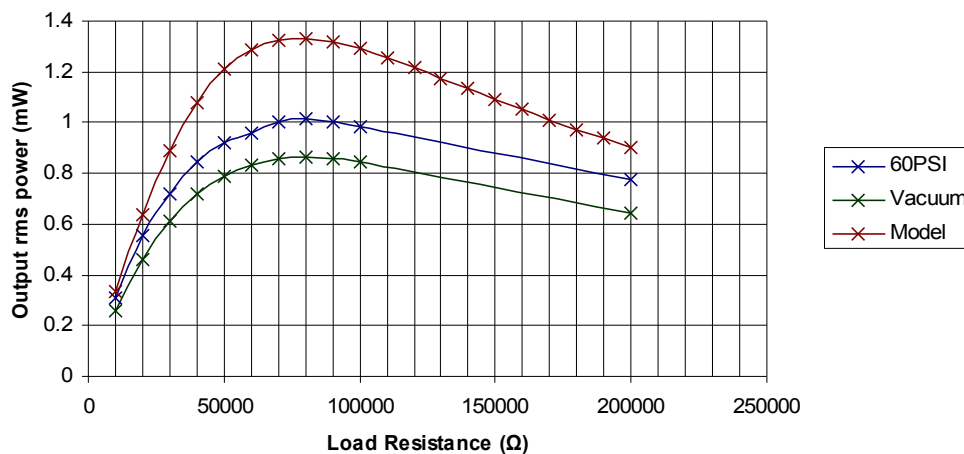
Graph 2 demonstrates how increasing the value of ϵ results in the peak power occurring at a lower load resistance, as inferred from equations 3.1 and 3.2. The values of ϵ estimated in literature ranged from 712 to 1694^[24, 25], but as can be seen from Graph 2, in order for the peak power to occur at 80kΩ as in the experiments, a relative permittivity of 2800 is required- much closer to PZTs permittivity of 3800 than expected. This is much higher than

the estimated values between 712 and 1694 from the literature, and suggests that the methods used for estimating MFC permittivity needs to be re-addressed. The mixing method is influenced by the structure of the electrodes, and as the electrodes only cover the PZT fibres rather than running continuously along both the PZT and the epoxy it is hypothesized that the mixing method results in an underestimate of the actual permittivity. This is an area that requires further research, and for the rest of the modelling a value of 2800 for the relative permittivity of MFC will be used.

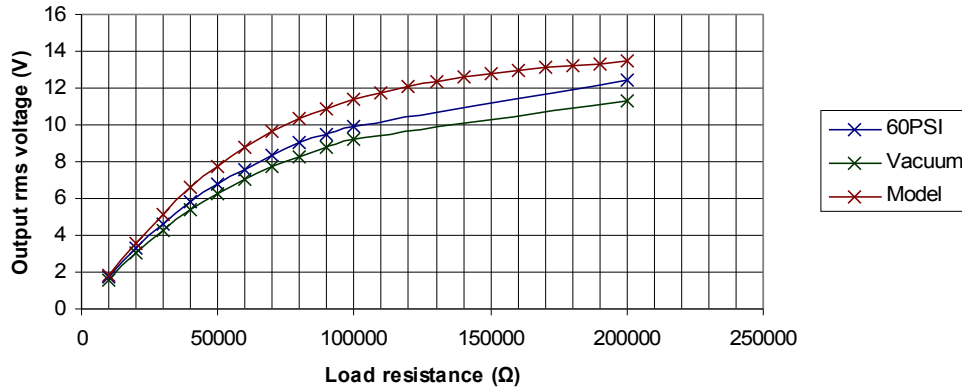
5. Presentation of Simulation Results

5.1. *Modelled Without Epoxy Bonding Layer*

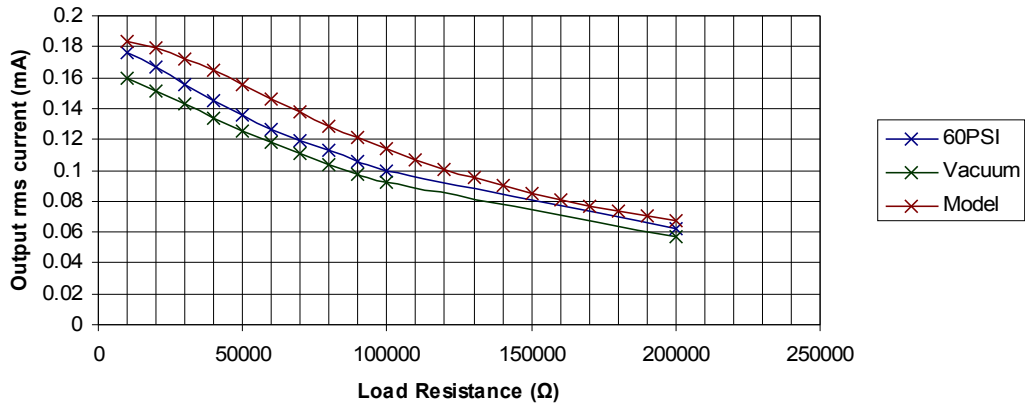
Graphs 3.1 - 3.3 display the results of the simulation in comparison to the experimental results. In the case of these 3 graphs, the simulation is run without an epoxy bonding layer modelled, the assumption made here is that the power loss due to the epoxy bonding layer is negligible. From the experimental results it is clear that this is not the case, however it is a simplification that allows the accuracy of the model to be easily compared to the experimental results. In the graphs the red line represents the model, and the blue and green lines represent the 60psi and vacuum cured experimental results respectively.



Graph 3.1: Simulated power output compared to experimental result



Graph 3.2: Simulated voltage output compared to experimental result



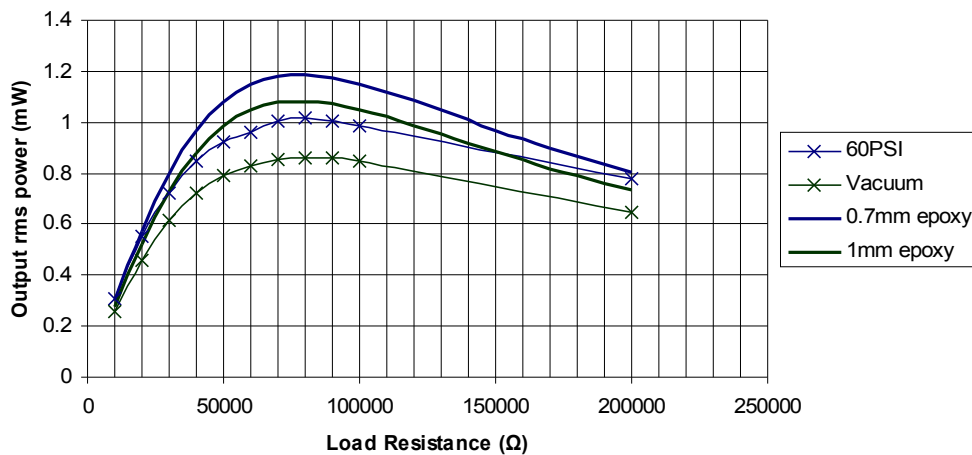
Graph 3.3: Simulated current output compared to experimental result

As the graphs show, the behaviour of the model closely matches the behaviour of the experimental EHDs discussed in part 4.1 of this report. The model shows a peak power output at 80kΩ, with the power output at $R_L=0$ and $R_L=\infty$ approaching zero.

The most glaring difference between the simulation and the experiment is the magnitude of the peak power. The peak power output of the 60psi experimental EHD under the stated conditions was 1.02mW, and the peak power of the simulation was 1.33mW, meaning the simulation predicts a power output 31% higher than the results obtained in the lab. This higher predicted power output can be explained however, as the model here doesn't take into account any energy losses. It is expected that significant energy losses should occur as heat energy in both the composite plate and MFC patch, and of course there are significant energy losses due to shear lag across the epoxy bonding layer that have not been modelled here.

5.2. Modelled Including Epoxy Bonding Layer

Graph 4 shows the power output results from the simulation when the epoxy bonding layer is included in the model, and compares this to the experimental results. In the graph the thick, unmarked blue and green lines represent the simulation results for a 0.7mm epoxy layer and 1mm epoxy layer respectively. 0.7mm is the epoxy layer thickness measured via a micrometer on the 60psi cured EHD, and 1mm the thickness measured on the vacuum cured EHD. The thin, marked lines represent the experimental results.



Graph 4: Simulation results compared with experimental results including the effect of shear lag across the epoxy bonding layer

As can be seen by comparing graph 3.1 with graph 4, the inclusion of the epoxy bonding layer in the model has decreased the power output of the simulation - as expected from the experimental results. It is clear however, that the simulated power output is still too high. The simulated power output is 0.175mW too high for the 60psi EHD, and 0.218mW too high for the vacuum EHD. These are significant differences, but it is difficult to tell if this is due to energy loss in the system, or if the model is over-estimating the power output of the devices. It does seem reasonable that a 0.2mW loss of power could be due to energy lost in the system but further research is required to confirm this.

Overall it seems that the model closely simulates the behaviour of the experimental EHDs.

6. Discussion and conclusions

The simulation results produced by the FEM closely match the results obtained from the experiment. The peak power outputs predicted by the model are higher than the experimental peak powers, but it is hypothesized that this is due to energy lost in the experimental system that was not taken into account in the model. The inclusion of the epoxy bonding layer in the model lowers the peak power of the model in a way similar to the effect of epoxy bonding layer thickness that was shown in the experiment.

The accuracy of the simulation shows that variations of this model could be used to design and optimise future MFC energy harvester designs for use in aircraft structural health monitoring and beyond. These would be able to power sensor nodes, negating the need for batteries and minimising the need for expensive aerospace maintenance work. Using the model to optimise the design means that additional money can be saved as the simulation can be run to ensure that the proposed designs are capable of fulfilling their requirements prior to manufacture.

Future research is required to validate the model over a range of different frequencies, as the only frequency tested in the experiment was 10Hz. This would enable designs to be optimised over a range of frequencies. Uncertainty surrounding the relative permittivity of MFC, and the permittivity analysis carried out in this report suggests that the mixing methods used in literature to estimate the value of the permittivity of MFC need to be re-addressed, more research should be carried out in this area to confirm this MFC property.

7. Project management, consideration of sustainability and health and safety

A Gantt chart was used to provide structure and aid time management for the duration of this project. The Initial Gantt chart that was formulated prior to the start of term 2 and present in the preliminary project report can be seen overleaf. The second Gantt chart is a revised version, created after delays in getting relevant signatures for the health and safety risk assessment forms resulted in beginning the experimental work later than anticipated. As well

as the provided Gantt charts a logbook was kept to provide structure to notes and ideas and to keep a record of the work carried out.

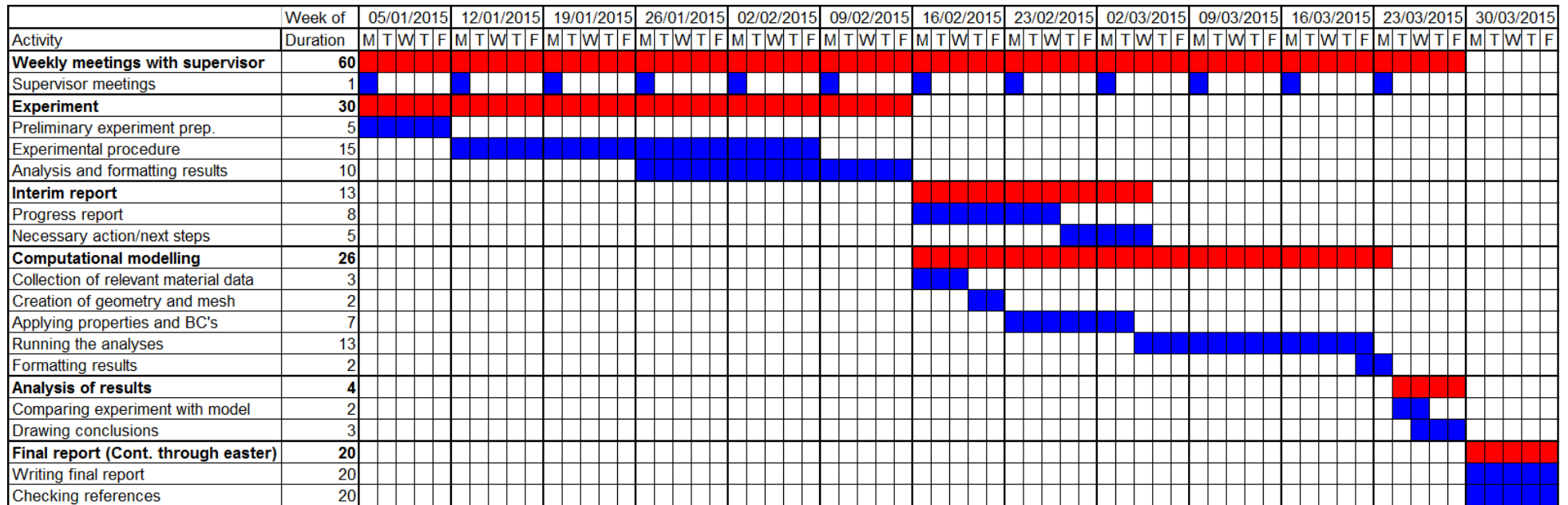
As part of the management of the project a risk assessment was completed in order to ensure that the project remained on track and that any risk or potential issues that could result in time delays or incompleteness of the project could be addressed and the risk minimised, this can also be found overleaf.

The main health and safety risks posed in this project came as a result of the experimental procedure. The use of the Instron testing machine and load resistance circuitry provided the source of the main risks, and as such a general health and safety risk assessment form was completed prior to carrying out the experimental procedure, as specified in the college's health and safety guidelines. The risk assessment outlines the risks and explains the measures in place to mitigate these risks and any further actions that needed to be taken to minimise the risks. This document can be found in the project logbook.

As well as the experimental health risk, modelling work requires long periods of time working at a computer. Because of this there are risks of computer related injuries such as back problems, eye strain, and repetitive strain injury. In order to minimise the risk of injury regular breaks were taken every hour to rest the eyes and joints, a comfortable chair was used to minimise back strain, and wrist rests and regular 5 minute typing breaks were employed to prevent RSI.

Assessing the sustainability of any engineering project is of the utmost importance. This project aims to advance the use of a technology that harvests energy from the ambient environment, with the potential to reduce the use of energy created by the burning of fossil fuels and contributing towards a more sustainable future. These potential benefits outweigh the use of energy required during the modelling and experimental phases of this project.

Project Plan - Term 2 - Start Date: 05/01/2015
Modelling macro-fibre composite energy harvesters for use in aircraft structural health monitoring



Gantt Chart 1: Initial project plan

Revised Project Plan - Term 2 - Start Date: 26/01/2015

Modelling macro-fibre composite energy harvesters for use in aircraft structural health monitoring

	Week of	26/01/2015	02/02/2015	09/02/2015	16/02/2015	23/02/2015	02/03/2015	09/03/2015	16/03/2015	23/03/2015	30/03/2015															
Activity	Duration	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Weekly meetings with supervisor	60																									
Supervisor meetings	1																									
Experiment	30																									
Preliminary experiment prep.	5																									
Experimental procedure	15																									
Analysis and formatting results	10																									
Interim report	13																									
Progress report	8																									
Necessary action/next steps	5																									
Computational modelling	26																									
Collection of relevant material data	3																									
Creation of geometry and mesh	2																									
Applying properties and BC's	7																									
Running the analyses	13																									
Formatting results	2																									
Analysis of results	4																									
Comparing experiment with model	2																									
Drawing conclusions	3																									
Final report (Cont. through easter)	20																									
Writing final report	20																									
Checking references	20																									

Gantt Chart 2: Revised project plan after delays

ID	Risk item	Effect	Cause	Likelihood	Severity	Importance	Action to minimise risk
1	Injury from test machine	Injury to myself or others in the lab resulting in incompletion of project	Crushing or ripping injuries from getting caught in the machine	L	S	S	Ensure that anyone using the machine is familiar with its operation, prevent access to the lab from unauthorised personnel, hands should not be placed in the test area.
2	Un-foreseen difficulty with the modelling	Additional time needed or unable to complete project	Unfamiliarity with ANSYS APDL	H	M	H	Study the application of APDL prior to modelling and plan time for mistakes
3	Un-foreseen delays in experimental procedure	Could cause delays in the completion of the project, or necessitate working longer hours to finish by the deadline	Unavailability of the lab, unexpected lack of materials or resources, health and safety issues etc.	L	M	M	Organise lab time well and communicate with supervisor about progress and any problems or issues. Fill out health and safety risk assessment swiftly.

Table 6: Project risk assessment

References

- [1] Victor Giurgiutiu (2008) "Structural health monitoring with piezoelectric wafer active sensors", Amsterdam: Academic Press.
- [2] Christian Boller, Wieslaw J. Staszewski (2004) "Structural Health Monitoring 2004", DEStech Publications, Inc.
- [3] Pozzi, M., Zhu, M. and Guo, S. (2012), "Harvesting energy from the dynamic deformation of an aircraft wing under gust loading", *Proc. SPIE 8348, Health Monitoring of Structural and Biological Systems*, vol. **8348**.
- [4] Person, M.R., Eaton, M. J., Pullen, R., Featherstone, C. A., Holford, K. M. (2012), "Energy Harvesting for Aerospace Structural Health Monitoring Systems", *Modern Practice in Stress and Vibration Analysis 2012, Journal of Physics: Conference Series*, vol. **382**.
- [5] Pozzi, M., Canziani, A., Durazo-Cardenas, I. and Zhu, M. (2012), "Experimental characterisation of macro fibre composites and monolithic piezoelectric transducers for strain energy harvesting", *Proceedings of the SPIE Health Monitoring of Structural and Biological Systems : Smart Structures/NDE 2012*, 12-15 March, San Diego, California, USA, Volume **8438**, pp.834832-1/10.
- [6] ANSYS. (2015). *ANSYS Mechanical*. Available: <http://www.ansys.com/Products/Simulation+Technology/Structural+Analysis/ANSYS+Mechanical>. Last accessed 21st Apr 2015.
- [7] Akyildiz, I. F., Su, W., Sankarasubramanianm, Y., Cayirci, E. (2002), "Wireless Sensor Networks: A Survey", *Computer Networks*, vol. **38**, pp. 393-422.
- [8] Boller, C, (2000). "Next generation structural health monitoring and its integration into aircraft design". *International journal of science*, vol. **31**, pp. 1333-1349.
- [9] Moss, S. Powlesland, I. Galea, S. Carman, G. (2010) "Vibro-impacting power harvester", *Proc. SPIE 7643, Active and Passive Smart Structures and Integrated Systems 2010*, 76431A (April 09, 2010)

- [10] Becker, T., Kluge, M., Schalk, J., Tiplady, K., Paget, C., Hilleringmann, U. and Otterpohl, a. T. (2009), "Autonomous Sensor Nodes for Aircraft Structural Health Monitoring", *IEEE Sensors Journal*, vol. **9**, no. 11, pp. 1589.
- [11] Penella López, M. T. and Gasulla Forner, M. (2007), "A review of commercial energy harvesters for autonomous sensors", *Instrumentation and Measurement Technology Conference – IMTC 2007 Warsaw, Poland, May 1-3, 2007*.
- [12] Samson, D., Kluge, M., Th. Becker and Schmid, U. (2011), "Wireless sensor node powered by aircraft specific thermoelectric energy harvesting", *Sensors & Actuators: A.Physical*, vol. **172**, no. 5-8, pp. 240-244.
- [13] Smart Material. (2003). *Macro Fibre Composite- MFC*. Available: <http://www.smart-material.com/MFC-product-main.html>. Last accessed 3rd April 2015.
- [14] Sodano, H, (2003). *Macro-Fiber Composites for Sensing, Actuation and Power Generation*. M.Sc.. Virginia: Virginia Polytechnic Institute and State University.
- [15] Guiliano, A., Marsic, M., Zhu, M. (2013), *Implementation and testing of a strain energy harvesting powered wireless sensor system for aircraft wing monitoring*. Green Computing and Communications (GreenCom) 2012. In *IEEE international conference*. Besancon, 20/11/2012. a: IEEE. 681 - 684
- [16] Zhu, M., Worthington, E. and Njuguna, J. (2009). *Analyses of Power Output of Piezoelectric Energy-Harvesting Devices Directly Connected to a Load Resistor Using a Coupled Piezoelectric-Circuit Finite Element Method*. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*. 56 (7), 1309-1318.
- [17] Yang, Y et al., (2009). *Vibration energy harvesting using macro-fiber composites*. Smart materials and Structures
- [18] National Instruments. (2015). *LabVIEW for Instrument Control*. Available: <http://www.ni.com/labview/applications/instrument-control/>. Last accessed 21st Apr 2015.
- [19] MathWorks. (2015). *MATLAB- The language of technical computing*. Available: <http://uk.mathworks.com/products/matlab/>. Last accessed 21st Apr 2015.
- [20] Elhajjar, R. Saponara, V. Muliana, A. (2014). *Smart Composites: Mechanics and Design*. Boca Raton, FL: Taylor and Francis. p146-147.
- [21] Huntsman. (2007). *Advanced materials: Araldite 2014-1*. Available: <http://www.intertronics.co.uk/data/ara2014-1.pdf>. Last accessed 12 Apr 2015.

- [22] Amir Javidinejad (2015). *Essentials of mechanical stress analysis*. Boca Raton, FL: Taylor and Francis. p10.
- [23] Cytec. (2015). *CYCOM ® 950 Epoxy Prepreg Technical Datasheet*. Available: http://cytec.com/sites/default/files/datasheets/CYCOM_950_041912.pdf. Last accessed 12th Apr 2015.
- [24] Giddings PF, et al, Modelling of piezoelectrically actuated bistable composites, *Mater Lett* (2011)
- [25] Yaowen Yang, Lihua Tang, Hongyun Li. (2009). Vibration energy harvesting using macro-fiber composites. *Smart materials and structures*. 18 (11)